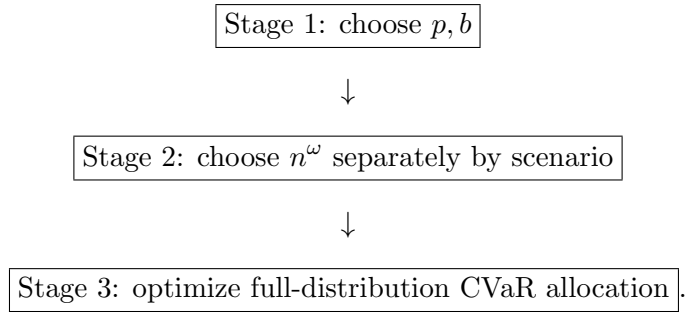


Staged Solution Method for the Type-Indexed Distance-State CVaR Model

1 Purpose and positioning

The monolithic stochastic mixed-integer program jointly chooses facility activations, vehicle bases, scenario-dependent vehicle routes, resource allocations, and CVaR variables. This provides the strongest optimization model, but it places all strategic and operational integer decisions for every scenario in one branch-and-bound tree.

The staged method is a decomposition-based matheuristic. It separates the strategic posture, detailed scenario routing, and final risk-aware allocation:



The method produces a feasible solution to the detailed model when all stages are feasible, but it does not generally provide a globally optimal solution because decisions fixed in an earlier stage cannot be revised later.

2 Compact monolithic reference model

Let

$$\begin{aligned}
 s &= (p, b), & \text{first-stage strategic decisions,} \\
 v^\omega &= (n^\omega, \bar{n}^\omega, x^\omega, y^\omega, z^\omega), & \text{scenario-}\omega \text{ recourse decisions.}
 \end{aligned}$$

Here p_i activates candidate site i , b_{kj} is the number of type- k vehicles based at j , and n_{kij}^ω is the integer count of type- k vehicles using arc (i, j) after consuming distance state d .

Let $\mathcal{S}$ denote the feasible strategic set containing site-count, activation-budget, fleet-assignment, and base-linking constraints. Let $\mathcal{F}_\omega(s)$ denote the detailed distance-state recourse set for scenario ω , including

- distance-expanded vehicle conservation;
- payload and degraded arc capacity;
- node-handling capacity;
- resource balance and unmet demand; and

- all variable domains.

The scenario loss is

$$\begin{aligned}\Lambda^\omega(s, v^\omega) = & \sum_{i \in N} \sum_{r \in R} \delta_{ir} z_{ir}^\omega \\ & + \sum_{k \in K} \sum_{(i,j) \in A_k} \sum_{r \in R} c_{kij} x_{kijr}^\omega \\ & + \varepsilon \sum_{k \in K} \sum_{(i,j) \in A_k} \sum_{d \in \mathcal{D}_{kij}} n_{kijd}^\omega.\end{aligned}\tag{1}$$

The monolithic reference formulation is

$$\min_{s, v, \eta, \xi} \quad \eta + \frac{1}{1 - \beta} \sum_{\omega \in \Omega} \pi^\omega \xi^\omega \tag{2}$$

$$\text{s.t.} \quad s \in \mathcal{S}, \tag{3}$$

$$v^\omega \in \mathcal{F}_\omega(s) \quad \forall \omega \in \Omega, \tag{4}$$

$$\xi^\omega \geq \Lambda^\omega(s, v^\omega) - \eta \quad \forall \omega \in \Omega, \tag{5}$$

$$\xi^\omega \geq 0 \quad \forall \omega \in \Omega, \quad \eta \in \mathbb{R}. \tag{6}$$

3 Stage 1A: reduced-scenario detailed strategic solve

This is the Stage 1 formulation used by the standard staged solver. Select a strategic scenario subset $\Omega_S \subseteq \Omega$. The probabilities used within the subset are

$$\hat{\pi}^\omega = \frac{\pi^\omega}{\sum_{\nu \in \Omega_S} \pi^\nu} \quad \forall \omega \in \Omega_S. \tag{7}$$

Stage 1A solves

$$\min_{s, v, \eta_S, \xi} \quad \eta_S + \frac{1}{1 - \beta} \sum_{\omega \in \Omega_S} \hat{\pi}^\omega \xi^\omega \tag{8}$$

$$\text{s.t.} \quad s \in \mathcal{S}, \tag{9}$$

$$v^\omega \in \mathcal{F}_\omega(s) \quad \forall \omega \in \Omega_S, \tag{10}$$

$$\xi^\omega \geq \Lambda^\omega(s, v^\omega) - \eta_S \quad \forall \omega \in \Omega_S, \tag{11}$$

$$\xi^\omega \geq 0 \quad \forall \omega \in \Omega_S, \quad \eta_S \in \mathbb{R}. \tag{12}$$

Let the resulting incumbent strategic posture be

$$s^H = (p^H, b^H). \tag{13}$$

The Stage 1 routes and allocations are used to value p and b , but they are discarded after the solve. Only p^H and b^H are passed to Stage 2.

If $\Omega_S = \Omega$, Stage 1A contains the complete monolithic formulation. The subsequent staged procedure nevertheless discards and reconstructs its operational decisions.

4 Optional Stage 1B: all-scenario strategic proxy

The proxy variant removes detailed vehicle-flow variables from Stage 1 and uses every scenario in Ω . It retains integer p and b but uses continuous direct-service variables:

$$\begin{aligned} t_{kji}^\omega &\geq 0, & \text{equivalent type-}k \text{ trips from base } j \text{ to demand node } i, \\ u_{kji}^\omega &\geq 0, & \text{resource } r \text{ delivered by those trips,} \\ u_{jr}^{\omega, \text{loc}} &\geq 0, & \text{resource } r \text{ used locally at activated site } j, \\ z_{ir}^\omega &\geq 0, & \text{proxy unmet demand.} \end{aligned}$$

For each feasible service option (ω, k, j, i) , precompute

$$\begin{aligned} L_{kji}^\omega &= \text{shortest-path mission distance,} \\ C_{kji}^\omega &= \text{minimum of vehicle payload and path bottleneck,} \\ \tilde{C}_{kji}^\omega &= \text{outbound unit transport cost.} \end{aligned}$$

An option is omitted if its path is unavailable or $L_{kji}^\omega > D_k$.

The proxy payload constraints are

$$\sum_{r \in R} w_r u_{kji}^\omega \leq C_{kji}^\omega t_{kji}^\omega \quad \forall (\omega, k, j, i) \text{ retained.} \quad (14)$$

Aggregate fleet-distance capacity is

$$\sum_{i: (\omega, k, j, i) \text{ retained}} L_{kji}^\omega t_{kji}^\omega \leq D_k a_j^\omega b_{kj} \quad \forall \omega, k, j. \quad (15)$$

For one-way proxy missions, each vehicle is used at most once:

$$\sum_{i: (\omega, k, j, i) \text{ retained}} t_{kji}^\omega \leq a_j^\omega b_{kj} \quad \forall \omega, k, j. \quad (16)$$

If return-to-base service is assumed, the return distance is included in L_{kji}^ω and constraint (??) is omitted, allowing repeated missions within the fleet-distance budget.

Inventory availability is

$$u_{jr}^{\omega, \text{loc}} + \sum_{k \in K} \sum_i u_{kji}^\omega \leq \rho I_{jr} a_j^\omega p_j \quad \forall \omega, j \in N^P, r. \quad (17)$$

Demand coverage is

$$\mathbb{I}_{\{i \in N^P\}} u_{ir}^{\omega, \text{loc}} + \sum_{k \in K} \sum_{j \in J_k} u_{kji}^\omega + z_{ir}^\omega = d_{ir}^\omega \quad \forall \omega, i, r. \quad (18)$$

The proxy also retains the scenario-degraded node-handling restriction

$$\sum_{k \in K_m} \sum_{j \in J_k} t_{kji}^\omega \leq \hat{\Theta}_{i,m}^\omega(p) \quad \forall \omega, m, i. \quad (19)$$

Proxy scenario loss is

$$\begin{aligned}\tilde{\Lambda}^\omega &= \sum_{i \in N} \sum_{r \in R} \delta_{ir} z_{ir}^\omega \\ &\quad + \sum_{k \in K} \sum_{j \in J_k} \sum_{i \in N} \sum_{r \in R} \tilde{c}_{kji}^\omega u_{kji}^\omega \\ &\quad + \varepsilon \sum_{k \in K} \sum_{j \in J_k} \sum_{i \in N} t_{kji}^\omega.\end{aligned}\tag{20}$$

Stage 1B solves

$$\min \quad \tilde{\eta} + \frac{1}{1-\beta} \sum_{\omega \in \Omega} \pi^\omega \tilde{\xi}^\omega \tag{21}$$

$$\text{s.t.} \quad s \in \mathcal{S}, \quad \text{constraints (??)–(??),}$$

$$\tilde{\xi}^\omega \geq \tilde{\Lambda}^\omega - \tilde{\eta} \quad \forall \omega \in \Omega, \tag{22}$$

$$\tilde{\xi}^\omega \geq 0 \quad \forall \omega \in \Omega, \quad \tilde{\eta} \geq 0. \tag{23}$$

Stage 1B minimizes CVaR of proxy losses, not detailed routed losses. Its output is again the strategic incumbent $s^H = (p^H, b^H)$.

5 Stage 2: independent detailed scenario routing

Fix the strategic posture obtained from Stage 1:

$$p = p^H, \quad b = b^H. \tag{24}$$

For every $\omega \in \Omega$, solve an independent detailed distance-state routing problem

$$Q_\omega(p^H, b^H) = \min_{v^\omega} \Lambda^\omega(p^H, b^H, v^\omega) \tag{25}$$

$$\text{s.t.} \quad v^\omega \in \mathcal{F}_\omega(p^H, b^H). \tag{26}$$

With one scenario, the CVaR expression reduces to that scenario's loss, so solving the one-scenario CVaR model is equivalent to equations (??)–(??).

Let the selected integer route be

$$n^{H,\omega} \quad \forall \omega \in \Omega. \tag{27}$$

The resource-allocation variables used to discover this route are discarded; they are reoptimized jointly in Stage 3. The scenario problems are mutually independent once p^H and b^H are fixed and may therefore be solved in parallel.

6 Stage 3: full-distribution CVaR allocation LP

Fix all strategic and route decisions:

$$p = p^H, \quad b = b^H, \quad n^\omega = n^{H,\omega} \quad \forall \omega \in \Omega. \tag{28}$$

The remaining decision variables are continuous resource allocations, terminations, unmet demand, retained resources, and CVaR variables. Define $\mathcal{X}_\omega(p^H, b^H, n^{H,\omega})$ as the continuous portion of the detailed recourse set. Stage 3 solves

$$z^H = \min_{x,y,z,\bar{n},\eta,\xi} \quad \eta + \frac{1}{1-\beta} \sum_{\omega \in \Omega} \pi^\omega \xi^\omega \quad (29)$$

$$\text{s.t.} \quad (x^\omega, y^\omega, z^\omega, \bar{n}^\omega) \in \mathcal{X}_\omega(p^H, b^H, n^{H,\omega}) \quad \forall \omega \in \Omega, \quad (30)$$

$$\xi^\omega \geq \Lambda^\omega(p^H, b^H, n^{H,\omega}, x^\omega, y^\omega, z^\omega) - \eta \quad \forall \omega \in \Omega, \quad (31)$$

$$\xi^\omega \geq 0 \quad \forall \omega \in \Omega, \quad \eta \in \mathbb{R}. \quad (32)$$

Because p , b , and n are fixed, Stage 3 is a linear program. Its value z^H is the best full-distribution CVaR attainable with the staged posture and routes.

7 Complete staged algorithm

Step 1: Generate the common scenario set Ω , probabilities π^ω , demands, severities, and degraded capacities.

Step 2: Solve either Stage 1A or Stage 1B and retain p^H, b^H .

Step 3: For each $\omega \in \Omega$, fix p^H, b^H , solve the detailed routing problem, and retain $n^{H,\omega}$.

Step 4: Build the all-scenario model, fix $p^H, b^H, n^{H,\omega}$, and solve the continuous CVaR allocation problem.

Step 5: Report z^H, η^H , scenario losses, tail scenarios, selected sites, vehicle base counts, routing gaps, and total runtime.

8 Feasibility, bounds, and interpretation

If every Stage 2 route is feasible and Stage 3 is feasible, the assembled solution is feasible for the detailed monolithic formulation. Since the problem is a minimization problem,

$$z^* \leq z^H, \quad (33)$$

where z^* is the unknown monolithic optimum. Thus z^H is a valid upper bound for the same detailed formulation and scenario distribution.

The staged method does not generate a global lower bound by itself. If LB is obtained from a valid LP relaxation or a monolithic branch-and-bound run, the staged solution may be assessed using

$$\text{Gap}_H = \frac{z^H - LB}{|z^H|} \times 100\%. \quad (34)$$

Stage 3 is conditionally optimal:

$$z^H = \min \{ \text{CVaR}_\beta(\Lambda^\omega) : p = p^H, b = b^H, n^\omega = n^{H,\omega} \}. \quad (35)$$

It is not generally globally optimal over all possible p , b , and n .

9 Comparison with the monolithic method

Feature	Monolithic method	Staged method
Strategic and operational decisions	Optimized jointly	Fixed sequentially
Scenario routing	One extensive-form MIP	One independent MIP per scenario
Final CVaR	Jointly determines all decisions	Reoptimizes allocation conditional on fixed decisions
Global lower bound	Produced by branch-and-bound	Requires a separate relaxation or monolithic run
Global optimality	Possible if solved to proof	Not guaranteed
Parallel opportunity	Primarily solver-internal	Stage 2 scenarios are independent
Operational output	Feasible incumbent if found	Feasible assembled solution if all stages succeed

10 Recommended reporting language

The method should be described as a *staged decomposition-based matheuristic* or *sequential fix-and-optimize method*. It should not be described as Benders decomposition because no feasibility or optimality cuts are returned to a strategic master problem. It should also not be described as exact unless the full monolithic formulation is solved to a valid global optimality certificate.

For a fair comparison between staged variants, compare the final Stage 3 objective values under the same scenarios, probabilities, β , distance buckets, and detailed routing formulation. Stage 1A and Stage 1B objective values are not directly comparable because they use different recourse models and may use different scenario sets.